

# Where Should Machine Learning Enter a Rule-Based Trading System? Matched-Budget Component Attribution Across Five Strategies and 153 Instruments

Author Name

**Abstract**—Hybrid trading systems combine a rule-based strategy with machine-learned components, yet evaluations rarely identify the decision at which machine learning (ML) adds value. We introduce matched-budget component attribution (MBCA), a controlled-variable protocol that holds data, features, model class, hyperparameters, freeze discipline, and evaluation fixed while replacing one insertion point at a time—regime filter, meta-labeling, position sizing, entry, or exit—with a learned counterpart. Study 1 applies MBCA to one strategy in three markets and finds 0 significant improvements and 20 significant losses in 60 configurations. Study 2, pre-registered in version control before any of its data existed, applies MBCA to five historically documented strategies on 153 daily series in 8 asset-class markets, with two independent freeze dates, close-price stop fills, per-market costs, mark-to-market portfolios, a paired block bootstrap, and false-discovery-rate control. Of 528 market-level tests, 99 have a raw  $p < 0.05$ , but only 8 survive as significant gains and 10 as significant losses. Narrowing components (meta-labeling, sizing) improve the Sharpe ratio in 67.6% and 70.7% of cells, are never significantly harmful, and transfer across markets, but their effects are small and replicate only partially across freezes. Replacing components are strategy dependent: ML entry is the most harmful component, whereas ML exit is the only one that significantly improves a strategy in several markets (RSI(2) mean reversion, 6 significant gains). Execution assumptions dominate every ML effect: stop-level fills inflate the controls’ Sharpe ratios by +0.34 to +0.77.

**Index Terms**—Algorithmic trading, backtest overfitting, block bootstrap, component attribution, false discovery rate, machine learning, meta-labeling, pre-registration, reproducibility, technical trading rules.

## I. INTRODUCTION

Machine learning (ML) has become a standard tool of empirical finance [1], and a common way to use it in trading is to attach a learned model to a rule-based strategy that is already understood and validated [2], [3]. The evidence offered for such hybrid systems is usually a single comparison: the hybrid against the rule-based baseline, summarized by one difference in Sharpe ratio. That comparison answers whether the hybrid performed better in one backtest. It does not answer the question a designer actually faces, namely *at which decision* a learned model should replace the rule.

Author Name is with the Department of [Department], [University], [City], [Country] (e-mail: author@example.com).

Code, data, and the pipeline that generates every table, figure, and number in this paper are available at [https://github.com/Cannonbold2412/ML\\_Component\\_Attribution](https://github.com/Cannonbold2412/ML_Component_Attribution).

A hybrid pipeline has several natural insertion points: whether a bar is admissible for trading (a regime filter), whether a fired signal should be taken (meta-labeling), how much to risk on it (position sizing), when to enter, and when to exit. An aggregate before-and-after comparison cannot attribute an improvement or a loss to any one of them. It is also confounded in ways that are easy to overlook: each learned component typically receives its own features, tuning budget, and evaluation window, so a difference between two hybrids can reflect feature engineering or evaluation leniency rather than the insertion point. Finally, the literature on technical trading rules shows that apparent performance is fragile to the number of configurations searched [4]–[6]; reporting the one hybrid that looked best compounds that problem.

This paper introduces *matched-budget component attribution* (MBCA), a controlled-variable protocol for this question. MBCA holds the data, the feature set, the model class and its hyperparameters, the freeze discipline, and the evaluation protocol fixed, and varies only which single decision of a validated rule-based system a learned model replaces. Because everything else is matched, a change in performance can be attributed to the insertion point itself, and a null result is a precise statement about the stated budget at that insertion point.

We apply MBCA twice. *Study 1* applies it to one validated strategy, a Jurik moving-average (JMA) crossover, in three markets and finds 0 significant improvements and 20 significant losses among 60 configurations; an independent re-implementation reproduces this headline but not its market-level claims. *Study 2* is a pre-registered test [7] of whether Study 1 generalizes. Its design was committed to version control before any of its data existed, and it evaluates five historically documented strategies on 153 daily series across 8 asset-class markets, at two independent freeze dates, under conservative execution, with mark-to-market portfolio accounting and a paired block-bootstrap test with false-discovery-rate (FDR) control.

The paper makes four contributions:

- 1) *Method*. MBCA and a taxonomy of insertion points (independent, narrowing, replacing) that predicts how much a component can change a rule-based system.
- 2) *Evidence*. 528 pre-registered market-level tests of five insertion points and one combination across five strategy families, with replication at a second freeze and a battery

of generalization, robustness, and bias checks.

- 3) *Auditability*. One configuration file drives data collection, every experiment, and every table, figure, and in-text number; a test suite enforces the methodological invariants and fails if the manuscript contains a hand-typed result.
- 4) *Software*. An open-source implementation with which practitioners can apply MBCA to their own strategy and data.

The evidence can be summarized in five findings. First, once multiplicity is controlled, ML rarely changes a historically successful rule significantly: 8 significant gains and 10 significant losses in 528 tests. Second, narrowing components (meta-labeling and position sizing) are the only ones that are directionally positive in most cells and never significantly harmful, and their effect transfers across markets; it is, however, small and replicates only partially across freezes. Third, replacing components are strategy dependent: ML entry is the most harmful component, whereas ML exit, which is negative on average for trend-followers, is the only component that significantly improves a strategy in several markets, in each case for RSI(2) mean reversion. Fourth, execution assumptions dominate: filling breached stops at the stop level rather than at the breaching close inflates the controls' Sharpe ratios by +0.34 to +0.77 and made ML exit look worse in Study 1 than it is. Fifth, the controls themselves lose most of their in-sample edge out of sample, which bounds what any component can add.

The remainder of the paper is organized as follows. Section II reviews related work. Section III defines MBCA and its taxonomy. Section IV reports Study 1. Sections V and VI present the design and results of Study 2. Section VII discusses the findings and their practical implications, Section VIII states the limitations, and Section IX concludes.

## II. RELATED WORK

### A. Machine Learning and Meta-Labeling in Trading

Gu, Kelly, and Xiu [1] show that tree ensembles and neural networks improve return forecasts and the portfolios built from them, largely by capturing nonlinear interactions among momentum, liquidity, and volatility signals. Their object is the forecast, not the decision it feeds. Meta-labeling [2] addresses the decision directly: a primary model chooses the side of a trade, and a secondary classifier, trained on realized outcomes, decides whether to take it and how large to make it. Joubert [3] formalizes meta-labeling as a layer that filters false-positive signals and sizes positions on top of a primary strategy. Our meta-labeling and position-sizing components implement exactly these two uses, with a fixed rule rather than a learned model as the primary; in the taxonomy of Section III-C both are *narrowing* components, because they can only reject or re-weight decisions the rule has already made.

### B. Hybrid Systems and Module Ablations

Hybrid systems that combine technical rules, ML, and other signals typically report aggregate improvements over a rule-only baseline without decomposing which sub-component

is responsible; a recent regime-adaptive equity system that integrates technical analysis, ML, and financial sentiment is a representative example [8]. The closest methodological precedent we found is a deep-learning multi-day-turnover pipeline for Chinese A-shares [9], which ablates its modules one at a time and reports each module's marginal contribution to annualized return. MBCA shares that spirit but differs in three ways: the components are insertion points of one fixed rule rather than modules of a proprietary architecture, every component receives the same feature and model budget, and every component is tested out of sample at a pre-specified freeze. Volatility-managed portfolios [10] scale exposure by realized volatility at the portfolio level; our sizing component instead scales each trade by a predicted probability of success.

### C. Technical Trading Rules and Data Snooping

Moving-average and breakout rules have a long empirical record [11], [12], and time-series momentum is documented across asset classes and more than a century of data [13], [14]. Their apparent profitability, however, is highly sensitive to how many rules were searched. White's reality check [5] and its application to technical rules by Sullivan, Timmermann, and White [4] remove much of it once the full rule universe is accounted for; Hansen's test of superior predictive ability [15] sharpens that comparison, and Harvey, Liu, and Zhu [6] argue for significance thresholds far above conventional ones. Study 2 therefore evaluates canonical rules chosen and parameterized before any data were seen, rather than searched ones.

### D. Inference on Sharpe Ratios and Backtest Overfitting

Sharpe ratios are estimated with substantial error, and serial correlation distorts naive annualization [16]. Ledoit and Wolf [17] propose a studentized test of the difference between two Sharpe ratios based on a circular block bootstrap [18], which respects both the pairing of the two return series and their serial dependence. Selection across many backtests inflates the best observed Sharpe ratio; the deflated Sharpe ratio [19] and the probability of backtest overfitting [20] quantify that bias. Across many tests, the false-discovery rate [21] and Holm's family-wise procedure [22] control multiplicity, and pre-registration [7] separates confirmatory tests from exploratory ones.

### E. Positioning

We found no work that holds a rule-based strategy, its data, features, model class, and evaluation protocol fixed while sweeping which single decision a learned model replaces, that names this constraint as a reusable methodology, and that reports every resulting configuration rather than the best one. MBCA is complementary to data-snooping corrections: those account for the search *across* rules, whereas MBCA controls what varies *within* one rule. This paper is the first full application of the methodology.

### III. MATCHED-BUDGET COMPONENT ATTRIBUTION

#### A. Setup and Notation

Consider a rule-based trading system applied bar by bar to instrument  $i$  at bar  $t$ . We decompose it into five decision functions,

$$\mathcal{S} = (g, e, f, s, x), \quad (1)$$

where the regime gate  $g$  decides whether a bar is admissible, the entry rule  $e \in \{-1, 0, +1\}$  fires a signal, the filter  $f$  decides whether a fired signal is taken, the sizing rule  $s$  sets the position size, and the exit rule  $x$  closes an open position. In a purely rule-based control,  $g$  admits every bar (or applies a rule such as an ADX threshold),  $f$  takes every signal, and  $s$  sizes every trade at one unit. A *component*  $c$  replaces exactly one of the five functions with a learned counterpart  $\hat{h}_c$ , giving the variant  $\mathcal{S}_c$ . The learned counterpart is trained under a budget

$$B = (\mathcal{D}, \mathbf{x}, \mathcal{H}, \lambda, \tau), \quad (2)$$

that is, training data  $\mathcal{D}$ , a feature map  $\mathbf{x}_{i,t}$ , a model class  $\mathcal{H}$  with hyperparameters  $\lambda$ , and a freeze date  $\tau$  after which the model is never refit. The quantity MBCA estimates for component  $c$  on an out-of-sample window  $W$  after  $\tau$  is

$$\Delta \text{SR}_c = \text{SR}(\mathcal{S}_c; W) - \text{SR}(\mathcal{S}; W), \quad (3)$$

where SR is the annualized Sharpe ratio of the system's daily portfolio returns.

#### B. The Matched Budget

MBCA holds five elements fixed across every tested insertion point: the *data* each model is trained and evaluated on; the *feature set* available to every learned component; the *model class and hyperparameters*, with no per-component tuning; the *training and freeze discipline*, under which each model is fit once and never refit on evaluation data; and the *evaluation protocol and statistical test* used to call a result significant. Only the insertion point varies between configurations (Table I), so a performance difference can be attributed to it rather than to feature engineering, tuning effort, or evaluation leniency. The shared budget used in both studies is listed in Table II. A negative result under MBCA is therefore a statement about this budget at this insertion point, not a claim that no learned model could help there.

#### C. A Taxonomy of Insertion Points

We classify each component by how much it can alter the set of trades the rule-based system would otherwise produce. An *independent* component gates admissibility orthogonally to the entry signal and can only reject bars; the ML regime filter is independent, and a rule-based ADX gate serves as its comparator. A *narrowing* component filters or re-weights decisions the rule has already made and cannot create trades; meta-labeling and position sizing are narrowing. A *replacing* component substitutes the rule's own mechanism and can produce a materially different trade set; learned entry and learned exit are replacing. The taxonomy implies an ex-ante hypothesis: because a narrowing component can at worst remove the rule's good trades, it bounds its own downside

and should produce smaller, lower-variance effects than a replacing component, whose effects can be large in either direction.

#### D. Components

Let  $\mathbf{x}_{i,t} \in \mathbb{R}^8$  be the shared, backward-looking feature vector of instrument  $i$  at bar  $t$  (Table II). Every classifier is a gradient-boosted tree ensemble [23] with the same hyperparameters, fit on standardized features.

- *ML regime filter*: entries are allowed iff  $P(\text{trend}_{i,t} = 1 \mid \mathbf{x}_{i,t}) \geq 0.5$ , where the label is one when the next 15 closes fit a straight line with  $R^2 \geq 0.5$  (linearity). Study 1 also tests two alternative label definitions (excursion and move size); Study 2 pre-registers the linearity label only, so it cannot be chosen after the fact.
- *ML meta-labeling*: at bars where the rule-based entry fires, the label is whether the trade the control would open is profitable; the trade is taken iff  $\hat{p}_{i,t}^{\text{meta}} \geq 0.5$  [2].
- *ML position sizing*: the same frozen meta-label model, consumed continuously; every trade is sized  $m_{i,t} = 2\hat{p}_{i,t}^{\text{meta}} \in [0, 2]$ , so  $\hat{p} = 0.5$  maps to one unit. No trade is added or removed.
- *ML entry*: two classifiers estimate whether a long (short) bracket entered at  $t$  with a 1.5 ATR stop and a 3 ATR target reaches the target first within 40 bars; the signal is the direction whose probability is larger and at least 0.5. This replaces the control's entry rule.
- *ML exit*: two classifiers estimate whether an open long (short) position is still favourable ten bars ahead; the position exits at the first bar at which that probability falls below 0.5, or at a 3 ATR safety stop. This replaces the control's exit rule.
- *Meta-label + sizing*: meta-labeling filters and sizing re-weights the surviving trades. In Study 1 this pairing was identified after the fact as the most promising; in Study 2 it was pre-registered, so its Study 2 result is an out-of-sample test of that choice.

#### E. Protocol: Walk-Forward Selection and Freezing

Each classifier is fit exactly once, on data before its freeze date; training rows whose label window crosses the freeze are excluded, and after the freeze the model only predicts. The parameters of the rule-based part of every pipeline (signal lengths and stop multiples) are re-selected by the same anchored walk-forward procedure for the control and for every variant: each instrument's history is split into five equal chronological folds, and the parameter set with the best mean in-sample trade return, subject to a minimum trade count, is chosen on fold  $k$  and traded on fold  $k + 1$ . Fig. 2 shows the resulting training and out-of-sample windows of both studies.

### IV. STUDY 1: ONE STRATEGY, THREE MARKETS

#### A. Design

The control is a JMA(7)/JMA(21) crossover whose parameters are walk-forward selected from a small grid. It exits with a fixed 1.5×ATR stop and 3×ATR target in forex, and with

TABLE I  
THE INSERTION POINTS TESTED, THEIR RULE-BASED CONTROLS, AND THEIR MATCHED-BUDGET LEARNED REPLACEMENTS

| Insertion point     | Rule-based control                              | Learned replacement                                     | Type        |
|---------------------|-------------------------------------------------|---------------------------------------------------------|-------------|
| Regime filter       | $ADX \geq 25$ gate (comparator), or no gate     | classifier of trending vs. choppy bars gates entries    | independent |
| Meta-labeling       | take every fired signal                         | take a fired trade iff $\hat{p}^{\text{meta}} \geq 0.5$ | narrowing   |
| Position sizing     | one unit per trade                              | size each trade by $2\hat{p}^{\text{meta}} \in [0, 2]$  | narrowing   |
| Entry               | rule signal (JMA, SMA, Donchian, TSMOM, RSI(2)) | long and short target-before-stop classifiers           | replacing   |
| Exit                | ATR stop and target, or trailing stop           | continuation classifier plus a 3 ATR safety stop        | replacing   |
| Meta-label + sizing | —                                               | meta-labeling filters, sizing re-weights the survivors  | combination |

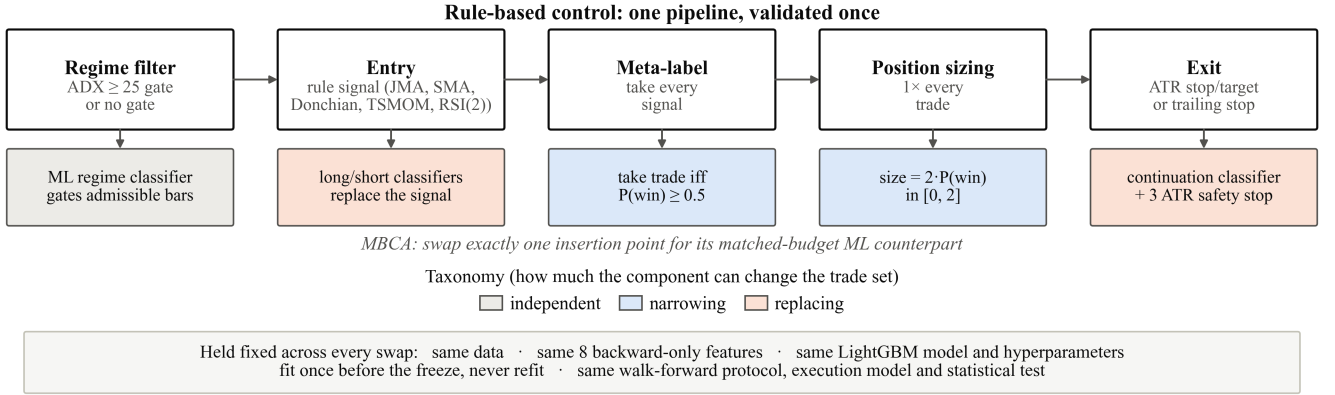


Fig. 1. The MBCA experiment. Each of the five decisions of one rule-based pipeline (top) is swapped, one at a time, for a learned counterpart (bottom, shaded by taxonomy), while the budget listed in the bottom band is held fixed across every swap.

TABLE II  
THE MATCHED BUDGET SHARED BY EVERY LEARNED COMPONENT (READ FROM THE CODE).

| Item                   | Setting                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Features               | ADX, ATR-normalized momentum, historical volatility, return $z$ -score, backward-looking $R^2$ , RSI, MACD histogram, KAMA slope (8, backward-only) |
| Model                  | LightGBM gradient-boosted trees on standardized features                                                                                            |
| n_estimators           | 200                                                                                                                                                 |
| num_leaves             | 31                                                                                                                                                  |
| max_depth              | 6                                                                                                                                                   |
| learning_rate          | 0.05                                                                                                                                                |
| min_child_samples      | 30                                                                                                                                                  |
| class_weight           | balanced                                                                                                                                            |
| random_state           | 42                                                                                                                                                  |
| Small model (ablation) | n_estimators = 50, num_leaves = 7, max_depth = 3                                                                                                    |

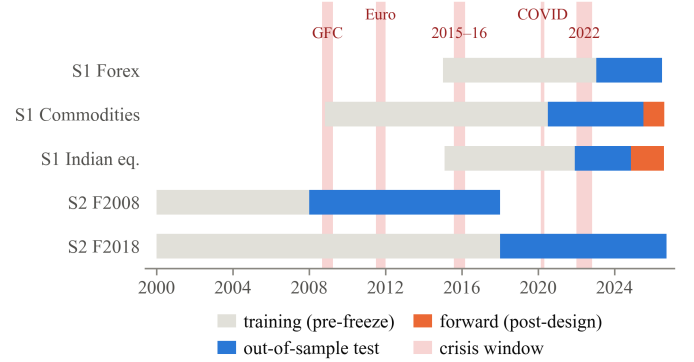


Fig. 2. Training and out-of-sample windows. Study 1 (S1) freezes each market at 70% of its calendar span and adds a forward window of data collected after the study was designed; Study 2 (S2) uses two independent freezes, F2008 and F2018. Shaded bands mark the five pre-registered crisis windows.

a  $1.5 \times \text{ATR}$  stop and a ratcheting trailing stop in commodities and Indian equities. The three markets are a four-pair forex sleeve (AUDNZD, CADJPY, AUDUSD, NZDUSD), five commodities (WTI and Brent crude, natural gas, gold, silver), and 48 NIFTY-50 stocks with an annual sector-curation overlay. Each market's models were frozen at 70% of its calendar span. Results are reported on a *holdout* window after the freeze and, for commodities and Indian equities, on a *forward* window of data collected after the study was designed (Fig. 2).

Study 1 keeps the statistical conventions of the original

study, which Appendix B states in full: each trade stream becomes a daily series by averaging the returns of the trades exiting each day, Sharpe ratios are annualized with a 6% per-year risk-free rate, and a configuration is called significant iff the 95% confidence interval (CI) of the per-day Sharpe difference, from 2,000 unpaired i.i.d. bootstrap resamples, excludes zero. The grid comprises five components, four combinations, three markets, and up to two windows, for 60 configurations in total.

## B. Results

Across the 60 configurations, 0 show a significant improvement and 20 a significant loss (Figs. 3 and 4). ML exit is the clearest failure: it loses significantly in 4 of its 5 configurations and inflates the trade count by a factor of 2.5–4.2, because a per-bar continuation classifier exits early and releases crossovers that a longer-held rule-based trade would have absorbed. On the original forex data, where the control’s holdout Sharpe ratio is 5.53, 10 of 12 configurations lose significantly. The closest call is meta-labeling followed by position sizing in Indian equities: the holdout Sharpe ratio rises from 2.84 to 4.01 (per-day CI  $[-0.050, 0.176]$ , approximately  $[-0.8, 2.8]$  annualized) and the forward Sharpe ratio from 1.05 to 2.70 (CI  $[-0.087, 0.293]$ ); neither change is significant. Every per-configuration value is provided as supplementary data (see the Data and Code Availability section).

## C. Independent Re-Implementation

The released package re-implements Study 1 from scratch (Appendix A). Running it on the bundled data is an independent replication with three known differences: a different forex data vendor with a flat 7bps cost, no sector-curation overlay, and a single walk-forward over the full history. Fig. 5 compares the two configuration by configuration. The re-implementation reproduces the headline: 0 significant gains, ML exit significantly harmful in every market with the trade count inflated by 2.4–3.0, and 49 of 60 configurations agreeing in sign. It does not reproduce the market-level claims. With different forex data (control Sharpe 1.39), only 2 forex configurations lose significantly, and meta-labeling flips from  $-4.19$  to  $+0.94$ . The meta-label + sizing near-miss replicates on the forward window ( $+1.65$  vs.  $+1.53$ ) but not on the holdout ( $+1.16$  vs.  $+0.04$ ), where the control itself rises to 4.31 without curation.

## D. Open Questions

Study 1 therefore leaves three questions open, which Study 2 was pre-registered to answer. Does the pattern hold for other historically successful strategies, not only a JMA crossover? Does it hold across a much broader and more diverse universe? And does it survive a second, independent freeze date, stricter execution assumptions, and a test that respects the pairing and serial dependence of returns?

## V. STUDY 2: DESIGN

### A. Pre-Registration and a Single Source of Truth

Every Study 2 design choice—universe, cleaning rules, strategies and parameter grids, components, freeze dates, execution model, statistics, regime definitions, and crisis windows—is declared in one configuration file that was committed to version control *before* any Study 2 data were downloaded or any result existed. The implementation reads nothing else: the same file drives data collection, every experiment, and every table, figure, and in-text number of this section and the next, all of which are generated by the pipeline rather than typed (Appendix A). The only deviation from the

TABLE III  
STUDY 2 UNIVERSE AFTER CLEANING (YAHOO FINANCE DAILY BARS TO 2026-09-21, CROSS-CHECKED AGAINST FRED AND MARKETWATCH). COST = ROUND-TRIP COST PLUS SLIPPAGE (BPS).

| Market         | N  | First bar  | Median bars | Cost | Surv.-free |
|----------------|----|------------|-------------|------|------------|
| Equity indices | 28 | 2000-01-03 | 6,654       | 5    | yes        |
| Equity ETFs    | 19 | 2000-01-03 | 6,719       | 5    | yes        |
| Fixed income   | 11 | 2000-09-21 | 6,075       | 5    | yes        |
| Commodities    | 19 | 2000-01-03 | 6,542       | 5    | yes        |
| FX             | 18 | 2000-01-03 | 5,929       | 3    | yes        |
| Crypto         | 4  | 2014-09-17 | 3,813       | 20   | no         |
| US stocks      | 20 | 2000-01-03 | 6,719       | 10   | no         |
| Intl stocks    | 34 | 2000-01-03 | 6,711       | 10   | no         |

pre-registration was forced by data availability and is logged: one planned symbol (Roche, ROG.SW) was not served by the data source, leaving 153 of 154 planned series. The regime-filter component was restricted in advance to a single label definition.

### B. Data and Provenance

Daily open, high, low, and close prices for 153 instruments in 8 markets (Table III, Fig. 6) were downloaded from the Yahoo Finance chart interface on 2026-09-21, covering 2000-01-03 to 2026-09-21 and 978,717 daily bars after cleaning. The universe comprises 28 equity indices from the Americas, Europe, and Asia-Pacific; 19 US sector and country ETFs; 11 bond ETFs and Treasury futures; 19 commodity futures; 18 currency pairs, including emerging-market currencies; 4 cryptocurrencies; and 20 US and 34 non-US large-capitalization stocks listed on 14 exchanges. The raw responses are stored byte for byte with their URL, UTC fetch time, and SHA-256 hash, and cleaning is a deterministic function of the raw bytes and the declared rules: split and dividend adjustment via the adjusted close, removal of 7,890 rows with missing or non-positive prices, 10,868 OHLC-consistency repairs, and 2 reversing bad ticks.

Two independent checks validate the data. Against 17 FRED series, five-day return correlations range from 0.82 to 1.00 and the median absolute price difference is at most 1.2%; daily FX correlations are lower only because the two sources fix FX at different times of day, and front-month natural-gas futures, a different instrument from Henry Hub spot, correlate at 0.64. A browser spot-check of 40 closing prices for 2 symbols against MarketWatch finds a maximum difference of 0.00.

### C. The Five Strategies

Five historically documented rule families serve as controls (Table IV): the JMA crossover of Study 1; the dual moving-average crossover of Brock, Lakonishok, and LeBaron [11]; the Donchian channel breakout of the Turtle rules [24]; time-series momentum [13], entered when the sign of the trailing six- or twelve-month return changes; and RSI(2) short-term mean reversion in the direction of the 200-day trend [25]. Four are trend-following and one is contrarian, and their holding periods range from days to months. Each control’s parameters are walk-forward selected per instrument exactly as in Study 1.

|             |                       | mean $\Delta$ Sharpe over windows (* = CI excludes 0) |             |                 | Recommendation (mechanical rule)                                               |
|-------------|-----------------------|-------------------------------------------------------|-------------|-----------------|--------------------------------------------------------------------------------|
| Type        |                       | Forex                                                 | Commodities | Indian equities |                                                                                |
| rule        | Rule ADX gate         | -1.02                                                 | -1.34       | -0.82           | Neutral: no consistent effect                                                  |
| independent | ML regime (linearity) | -5.20*                                                | -0.61       | -1.33           | Avoid: significant loss in Forex                                               |
| independent | ML regime (excursion) | -4.66*                                                | +0.23       | -0.76           | Mixed: avoid in Forex; promising, unreplicated in Commodities                  |
| independent | ML regime (move size) | -4.78*                                                | -4.65*      | -0.84           | Avoid: significant loss in Forex, Commodities                                  |
| narrowing   | ML meta-labeling      | -4.19*                                                | +0.75       | +1.44           | Mixed: avoid in Forex; promising, unreplicated in Commodities, Indian equities |
| narrowing   | ML position sizing    | -1.03                                                 | -0.44       | +0.49           | Worth exploring in Indian equities: unreplicated                               |
| replacing   | ML entry              | -4.46*                                                | -1.61       | +0.36           | Mixed: avoid in Forex; promising, unreplicated in Indian equities              |
| replacing   | ML exit               | -6.48*                                                | -4.84*      | -1.49*          | Avoid: significant loss in Forex, Commodities, Indian equities                 |
| combination | Entry + exit          | -4.88*                                                | -4.49*      | -2.65*          | Avoid: significant loss in Forex, Commodities, Indian equities                 |
| combination | Entry + sizing        | -3.99*                                                | -2.44       | +0.50           | Mixed: avoid in Forex; promising, unreplicated in Indian equities              |
| combination | Exit + sizing         | -6.21*                                                | -4.43*      | -0.90*          | Avoid: significant loss in Forex, Commodities, Indian equities                 |
| combination | Meta-label + sizing   | -4.27*                                                | +0.55       | +1.40           | Mixed: avoid in Forex; promising, unreplicated in Commodities, Indian equities |

Fig. 3. Study 1 opportunity matrix: mean  $\Delta$ Sharpe over windows for each component and market (\* = CI excludes zero in at least one window), with the taxonomy type and the recommendation produced by the mechanical rule of the original study.

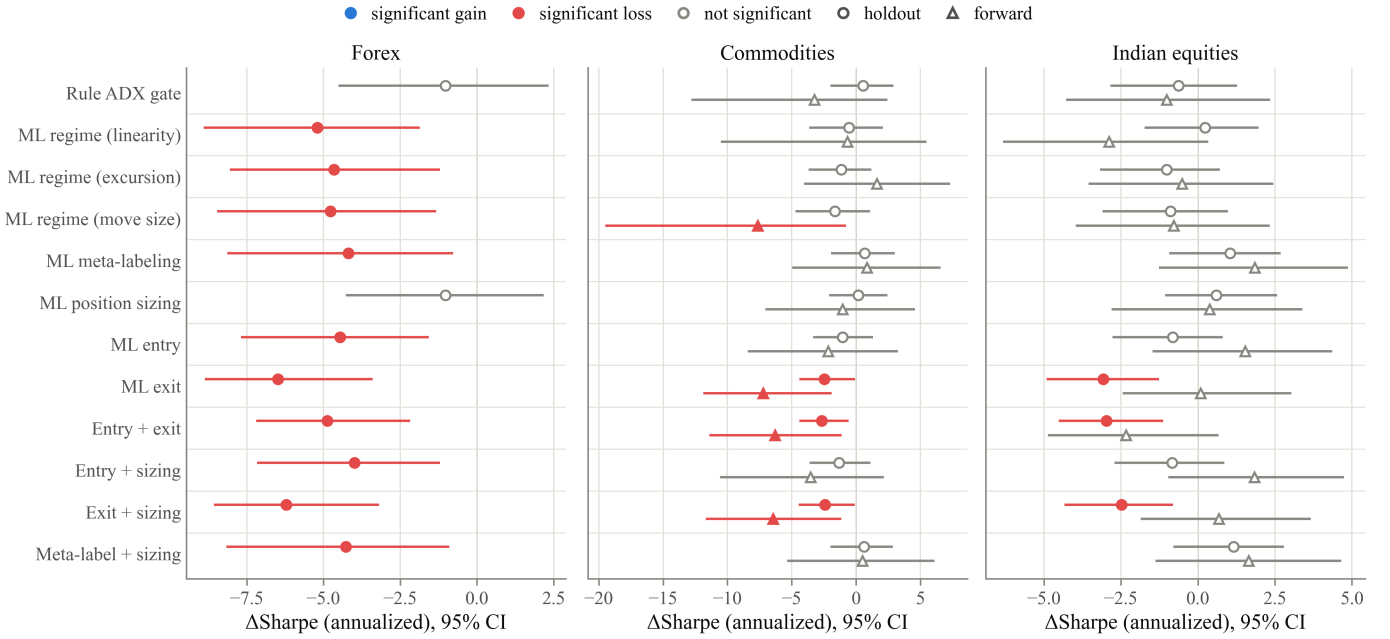


Fig. 4. Every Study 1 configuration. Markers show the annualized  $\Delta$ Sharpe (circles: holdout; triangles: forward) and bars the bootstrap CI rescaled by  $\sqrt{252}$  so that both share one axis. Filled markers are significant; the forex market has no forward window.

TABLE IV

THE FIVE PRE-REGISTERED STRATEGIES (THE RULE-BASED CONTROLS). EACH INSTRUMENT'S PARAMETERS ARE RE-SELECTED FROM THESE GRIDS BY WALK-FORWARD ON EVERY FOLD; EXIT MULTIPLES ARE IN UNITS OF ATR.

| Strategy             | Reference       | Signal grid                                     | Exit     | Exit grid ( $\times$ ATR)                     |
|----------------------|-----------------|-------------------------------------------------|----------|-----------------------------------------------|
| JMA crossover        | Study 1 control | fast $\in \{5, 7, 9\}$ ; slow $\in \{21, 28\}$  | trailing | stop $\in \{1.5, 2\}$ ; trail $\in \{2, 3\}$  |
| Dual SMA crossover   | [11]            | fast $\in \{20, 50\}$ ; slow $\in \{100, 200\}$ | trailing | stop $\in \{2, 3\}$ ; trail $\in \{3, 4\}$    |
| Donchian breakout    | [24]            | lookback $\in \{20, 55\}$                       | trailing | stop $\in \{2\}$ ; trail $\in \{2, 3\}$       |
| Time-series momentum | [13]            | lookback $\in \{126, 252\}$                     | trailing | stop $\in \{2, 3\}$ ; trail $\in \{3, 4\}$    |
| RSI(2) reversion     | [25]            | entry $\in \{5, 10\}$                           | bracket  | stop $\in \{2, 3\}$ ; target $\in \{1, 1.5\}$ |

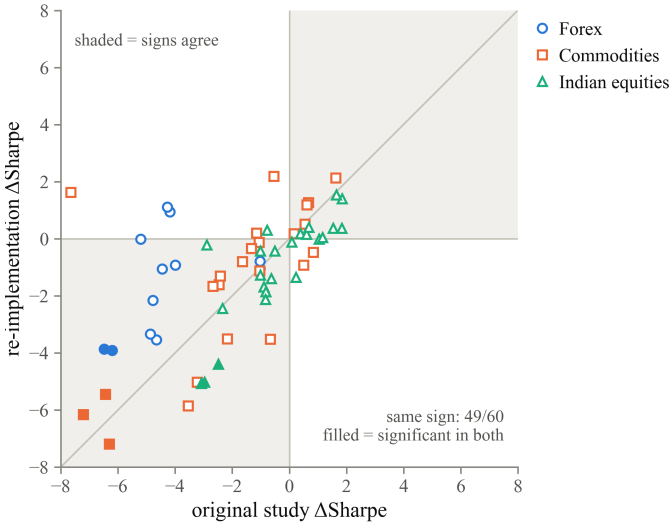


Fig. 5. Study 1: original  $\Delta$ Sharpe against the independent re-implementation, one point per configuration. Shaded quadrants mark sign agreement; filled markers are significant in both.

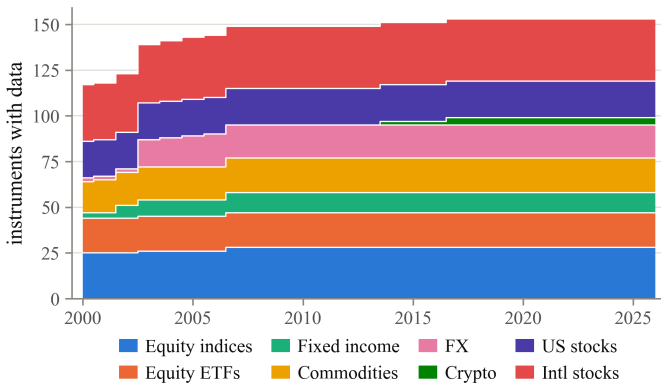


Fig. 6. Study 2 universe: number of instruments with data in each year, by market. Crypto enters in 2014; single stocks are current large caps and therefore survivorship-prone (Table III).

#### D. Execution and Portfolio Accounting

Execution is deliberately conservative. Signals and exits are evaluated on daily closes; a stop breached by a close is filled *at that close*, not at the stop level (the optimistic stop-level fill of Study 1 is re-run as a sensitivity), and entries fill at the signal bar's close (a one-day entry lag is re-run as a sensitivity). The round-trip transaction cost plus slippage is set per market in advance—3 bps for FX, 5 bps for indices, ETFs, and futures, 10 bps for single stocks, and 20 bps for crypto—and varied by the multipliers  $\times 0$ ,  $\times 2$ ,  $\times 4$ . Performance is measured on *mark-to-market* daily portfolios: capital is split equally across the instruments of a market (or of the whole universe) that trade on a given day, each position earns its instrument's close-to-close return from entry to exit, and half of the round-trip cost is charged at entry and half at exit, scaled by position size. Sharpe and Sortino ratios use no risk-free rate, because the universe spans many currencies.

#### E. Windows, Generalization, and Robustness Design

Two independent freezes give two non-overlapping out-of-sample windows, each with its own frozen models (Fig. 2): F2008, with models trained on 2000–2007 and tested on 2008–2017, a window that includes the global financial crisis; and F2018, trained on 2000–2017 and tested from 2018 onward, including the COVID crash and the 2022 bear market. Crypto enters only F2018. Each (strategy, market, freeze) comparison is a *market-level* test, and the equal-weight portfolio of all 153 instruments gives a *global* test per (strategy, freeze).

Beyond the main grid, the design pre-registers the following checks: *cross-market transfer*, in which meta-labeling and meta-label + sizing models are fit on every other market and applied to the held-out one; *budget ablations*, with meta-label thresholds of 0.45 and 0.55 and a small model of 50 trees of depth 3 for meta-labeling and exit; *execution* and *transaction-cost* sensitivities; *cross-regime* tests, with bull and bear regimes defined by the market index's position relative to its 200-day average and high- and low-volatility regimes by 20-day realized volatility relative to its expanding median, both known at  $t - 1$ ; five crisis windows; a *survivorship* split between survivorship-free indices, ETFs, futures, and FX and current-constituent stocks and surviving coins; leave-one-instrument-out and leave-one-market-out jackknives; and an in-sample versus out-of-sample decay check of the controls' parameter selection.

#### F. Inference

The difference in annualized Sharpe ratio between a variant and its control is tested with a paired circular-block bootstrap [17], [18]. Both daily return series are aligned on the same dates, and the *same* 20-day blocks are resampled for both in each of 2,000 replicates, which preserves their cross-correlation and short-range serial dependence and corrects the unpaired i.i.d. test of Study 1. The two-sided  $p$ -value is  $2 \min\{P(\Delta^* \leq 0), P(\Delta^* \geq 0)\}$ , where  $\Delta^*$  is the bootstrap replicate of  $\Delta$ SR. Multiplicity is controlled with the Benjamini–Hochberg FDR [21] within pre-defined families (the 528 market-level main tests, the 70 global main tests, and the transfer, ablation, and execution families), with Holm's method [22] reported as a family-wise check in the spirit of [4], [6]. A difference is called significant iff  $q < 0.05$ . Deflated Sharpe ratios [19] are reported for every portfolio in the supplementary data.

### VI. STUDY 2: RESULTS

#### A. Headline: Few Significant Effects After Multiplicity Control

The pre-registered grid yields 528 market-level main tests (seven components, five strategies, eight markets, and two freezes, less the cells without training data). Of these, 99 have a raw  $p < 0.05$ , far more than the roughly 26 expected under a global null (Fig. 7): the components do change performance. After Benjamini–Hochberg correction, however, only 8 are significant gains and 10 significant losses (Holm: 16 of either sign), and 54.9% of all market-level differences are positive.

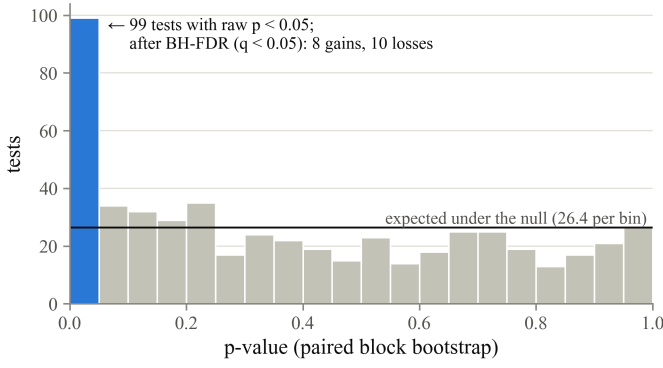


Fig. 7. Distribution of the 528 market-level  $p$ -values. The excess in the first bin shows that the components change performance; FDR control leaves few of those changes significant.

On the all-market portfolio, which diversifies across 153 instruments, 1 of the 70 global tests is a significant gain and 1 a significant loss. No component significantly improves any of the four trend-following strategies on the all-market portfolio in either freeze.

### B. Component Attribution

Table V and Figs. 8 and 9 give the per-component picture, and Table VI breaks it down by strategy. Four regularities refine the taxonomy of Section III-C.

*Narrowing components are the most consistently positive and never significantly harmful.* Meta-labeling improves the Sharpe ratio in 67.6% of market-level cells (median +0.12; 1 significant gain, 0 significant losses), meta-label + sizing in 67.6% (median +0.12), and position sizing in 70.7% (median +0.05, with no significant result in either direction). Both also cut turnover, by a median of -3.46 and -1.90 capital per year. Their gains are small relative to their uncertainty.

*ML entry is the most harmful component.* It loses in most cells (median -0.15; 8 significant losses) and raises turnover by +10.40 capital per year. The damage is concentrated in the contrarian strategy: 7 of its significant losses are RSI(2) cells (median -1.07), where a learned, trend-oriented entry replaces a mean-reversion rule.

*ML exit is strategy dependent rather than uniformly harmful.* For the four trend-following controls its median effect is -0.15, but for RSI(2) it is +0.70, with 6 significant gains—the only component that significantly improves a strategy in several markets (all-market portfolio, F2008: +0.78, CI [+0.49, +1.12]). The RSI(2) rule’s fixed ATR profit target caps winners, and a learned continuation exit lets them run; for trend-followers, whose trailing stops already let winners run, the learned exit mostly adds turnover (median +5.36).

*Independent gates do nothing reliable.* The ML regime filter (median -0.01, 0 significant results) and the rule-based ADX gate (median -0.01, 1 significant loss) are indistinguishable from zero.

### C. The All-Market Portfolio

Fig. 10 reports every global before-and-after comparison with its paired bootstrap CI, Fig. 11 the corresponding equity

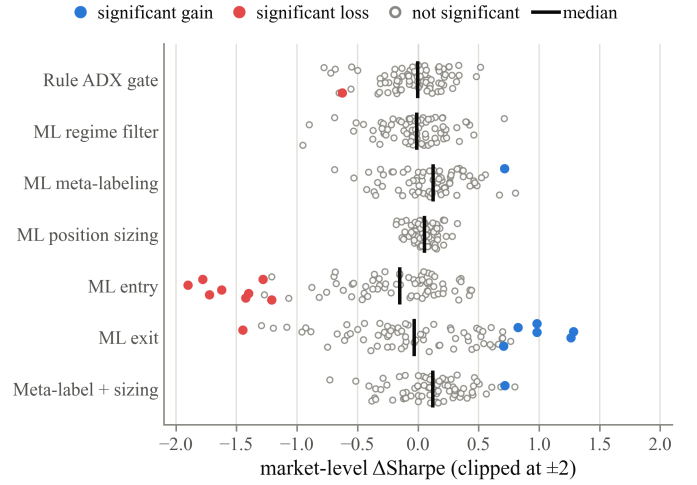


Fig. 8. Every market-level  $\Delta$ Sharpe, by component. Filled markers are significant at  $q < 0.05$ ; vertical bars mark the median.

curves, and Fig. 12 the change in the full risk profile. Two facts frame all ML effects. First, the rule-based controls are weak out of sample: their all-market Sharpe ratios range from -0.39 (RSI(2), F2008) to 0.69 (JMA, F2008). Second, most drawdown changes are mechanical exposure effects rather than skill. Every component that trades less makes the all-market maximum drawdown shallower in most cases (meta-labeling 10/10, meta-label + sizing 9/10, ML exit 9/10, ML regime filter 9/10; median exposure change -13.1%, -8.1%, -20.2%, and -9.8%), whereas ML entry, which raises exposure by +56.7%, deepens it in every case (0/10 shallower). The risk-adjusted return, not the drawdown, is therefore the right yardstick for attribution.

### D. Replication Across Freezes

The two freezes are independent replications with different training data and non-overlapping test periods (Fig. 13); only cells observed in both freezes are compared. Agreement is modest. The sign of a strategy  $\times$  market effect agrees across freezes in 74.3% of cells for meta-labeling, 74.3% for meta-label + sizing, 74.3% for ML exit, and 74.3% for ML entry, but only 51.4% and 54.3%—close to chance—for the regime filter and position sizing. Magnitudes separate the components more clearly: the cross-freeze correlation of  $\Delta$ Sharpe is 0.70 for exit and 0.69 for entry, but 0.48 for meta-labeling and 0.14 for sizing. The components with large effects replicate best; the small positive effects of the narrowing components replicate in sign more often than in size.

### E. Transfer, Budget, and Execution

Fig. 14 collects three pre-registered robustness checks. *Cross-market transfer* (panel a): models fit on every other market and applied to a held-out one do at least as well as models fit in-market. Transferred meta-labeling has a median  $\Delta$ Sharpe of +0.16 (positive in 65.4% of cells, 0 significant losses) and transferred meta-label + sizing +0.15 (65.4%), against +0.12 and +0.12 in-market, so whatever the meta-label

TABLE V

COMPONENT ATTRIBUTION ACROSS THE UNIVERSE. MARKET-LEVEL TESTS: EVERY STRATEGY  $\times$  MARKET  $\times$  FREEZE. SIG.  $+/-$  = BH-FDR  $q < 0.05$  WITHIN THE FAMILY. GLOBAL = ALL-MARKET PORTFOLIO (STRATEGY  $\times$  FREEZE). FREEZE AGREEMENT = SHARE OF STRATEGY  $\times$  MARKET CELLS WHOSE  $\Delta$ SR HAS THE SAME SIGN IN BOTH INDEPENDENT FREEZES.  $\Delta$ TUR. IN CAPITAL/YEAR.

| Component           | Type        | Tests | $\Delta$ SR > 0 | Median $\Delta$ SR | Sig.+ | Sig.- | Global median | Global > 0 | Freeze agree | $\Delta$ Tur. |
|---------------------|-------------|-------|-----------------|--------------------|-------|-------|---------------|------------|--------------|---------------|
| Rule ADX gate       | rule        | 80    | 46.2%           | -0.01              | 0     | 1     | -0.02         | 5/10       | 60.0%        | -2.36         |
| ML regime filter    | independent | 75    | 45.3%           | -0.01              | 0     | 0     | -0.02         | 5/10       | 51.4%        | -3.39         |
| ML meta-labeling    | narrowing   | 74    | 67.6%           | +0.12              | 1     | 0     | +0.08         | 6/10       | 74.3%        | -4.42         |
| ML position sizing  | narrowing   | 75    | 70.7%           | +0.05              | 0     | 0     | +0.03         | 6/10       | 54.3%        | -1.67         |
| ML entry            | replacing   | 75    | 40.0%           | -0.15              | 0     | 8     | -0.24         | 2/10       | 74.3%        | +9.54         |
| ML exit             | replacing   | 75    | 48.0%           | -0.03              | 6     | 1     | -0.17         | 3/10       | 74.3%        | +4.87         |
| Meta-label + sizing | combination | 74    | 67.6%           | +0.12              | 1     | 0     | +0.08         | 6/10       | 74.3%        | -2.79         |

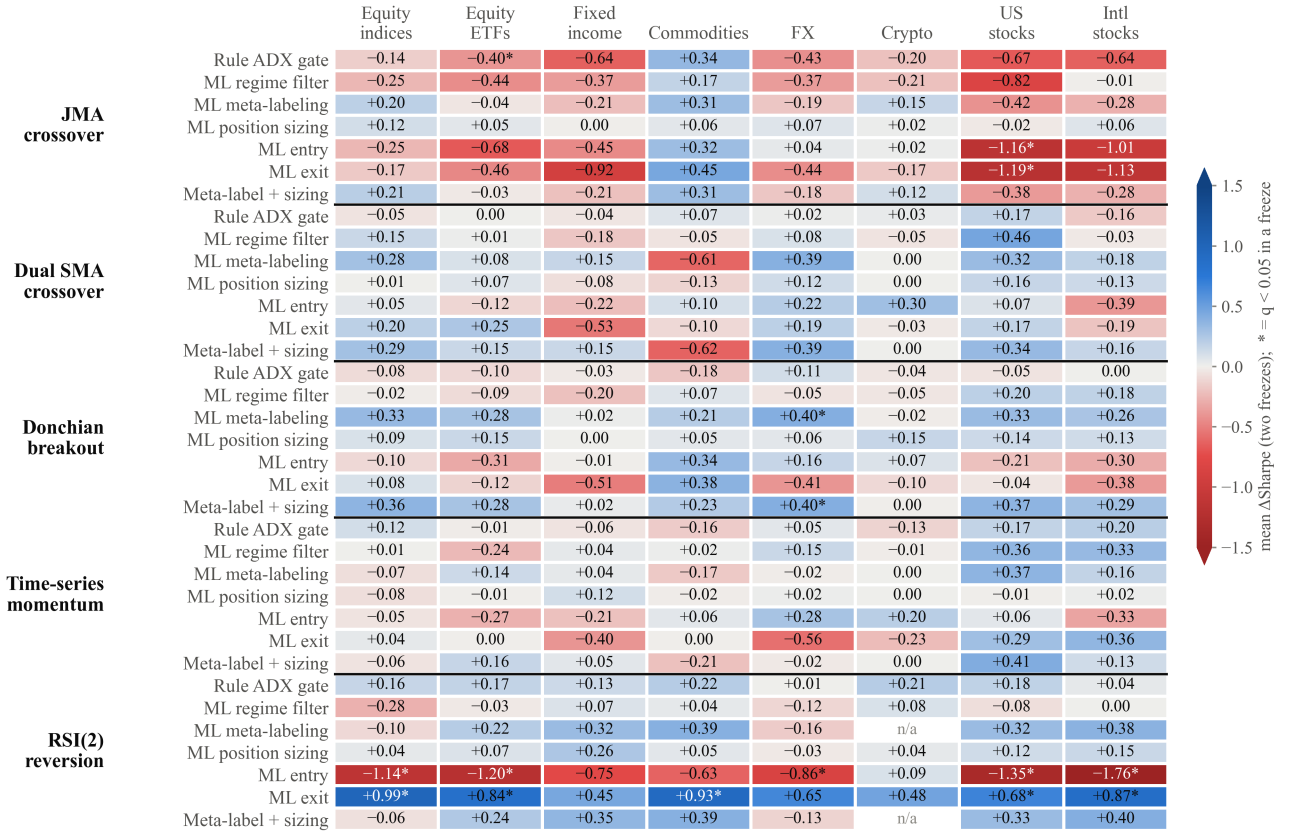


Fig. 9. Where ML changes each strategy: mean  $\Delta$ Sharpe over the two freezes for every strategy, component, and market (\* = significant at  $q < 0.05$  in at least one freeze; n/a = no training data). The ML exit and ML entry rows of RSI(2) reversion carry most of the significant cells.

TABLE VI

MEDIAN MARKET-LEVEL  $\Delta$ SR BY STRATEGY AND COMPONENT (SIGNIFICANT GAINS / SIGNIFICANT LOSSES AT  $q < 0.05$  IN PARENTHESES).

| Strategy             | ADX gate    | ML regime   | ML meta     | ML sizing   | ML entry    | ML exit     | Meta+sizing |
|----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| JMA crossover        | -0.42 (0/1) | -0.28 (0/0) | -0.12 (0/0) | +0.05 (0/0) | -0.28 (0/1) | -0.59 (0/1) | -0.11 (0/0) |
| Dual SMA crossover   | +0.01 (0/0) | -0.01 (0/0) | +0.21 (0/0) | +0.04 (0/0) | +0.06 (0/0) | -0.08 (0/0) | +0.21 (0/0) |
| Donchian breakout    | -0.04 (0/0) | +0.00 (0/0) | +0.18 (1/0) | +0.10 (0/0) | -0.06 (0/0) | -0.14 (0/0) | +0.22 (1/0) |
| Time-series momentum | +0.03 (0/0) | +0.06 (0/0) | +0.09 (0/0) | +0.00 (0/0) | +0.00 (0/0) | -0.11 (0/0) | +0.10 (0/0) |
| RSI(2) reversion     | +0.14 (0/0) | -0.06 (0/0) | +0.20 (0/0) | +0.08 (0/0) | -1.07 (0/7) | +0.70 (6/0) | +0.21 (0/0) |

model learns is not market specific. *Budget ablations* (panel b): moving the meta-label threshold to 0.45 or 0.55 (median +0.06 and +0.08) or shrinking the model to 50 trees of depth 3 (meta-labeling +0.19, exit -0.07) leaves the conclusions unchanged; the effects come from the insertion point, not from

the particular budget.

*Execution assumptions* (panel c) matter more than any ML component. Filling a breached stop at the stop level instead of at the breaching close—the Study 1 convention—raises the rule-based control’s all-market Sharpe ratio by +0.34 to

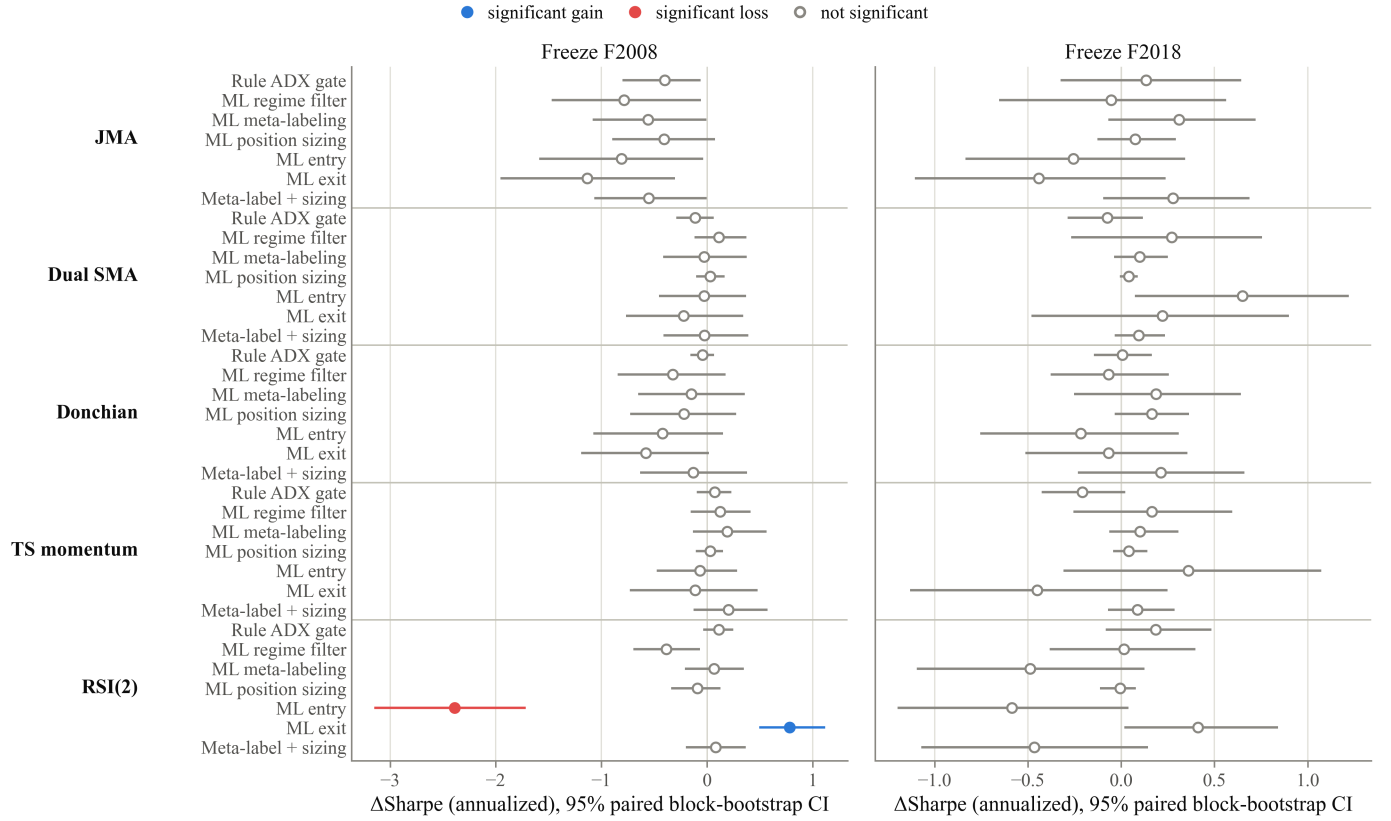


Fig. 10. All-market portfolio:  $\Delta$ Sharpe with 95% paired block-bootstrap CIs for every strategy and component, in each freeze. Filled markers are significant at  $q < 0.05$  across the 70 global tests.

+0.77 (significant in 10/10 strategy  $\times$  freeze cells). Under that optimistic fill, ML exit looks far worse (median -0.52, 5/10 significant) than under realistic fills (median -0.17), because the fill flatters the control, not the exit. Part of Study 1's finding that ML exit is harmful is therefore an artifact of optimistic stop fills. A one-day entry lag changes the control by a median of -0.04 and leaves the ML comparisons qualitatively unchanged.

#### F. Transaction Costs

The narrowing components are insensitive to transaction costs because they reduce turnover (Fig. 15): the median all-market  $\Delta$ Sharpe of meta-label + sizing is +0.09, +0.08, +0.08, and +0.06 at cost multipliers of zero, one, two, and four. ML exit, which raises turnover, decays from +0.02 to -0.89 over the same range.

#### G. Regimes and Crises

Narrowing components help in low-volatility regimes (meta-labeling +0.38, meta-label + sizing +0.39) and slightly hurt in high-volatility regimes (-0.12 and -0.12); ML exit helps in bear markets (+0.38) and hurts in bull markets (-0.42), consistent with cutting trades early (Fig. 16a). The crisis windows show the exposure pattern of Fig. 12 again: every component except ML entry makes the drawdown shallower in each of the five crises, whereas ML entry, the only component that raises exposure, deepens it in each (Fig. 16c).

#### H. Fragility and Parameter Decay

Effects in survivorship-free markets and in current-constituent stocks and crypto are similar for the narrowing components (meta-label + sizing median +0.11 vs. +0.14; meta-labeling +0.10 vs. +0.16) and small for exit (+0.03 vs. -0.11), so survivorship does not drive the conclusions. Dropping any single instrument flips the sign of 24.6% of market-level effects, so most of them are fragile, but none of the 8+10 significant ones (0/18 flip; Fig. 17); dropping any single market flips 5.7% of the global effects.

The controls' own parameter selection decays sharply out of sample (Fig. 18). The dual SMA crossover earns 330 bps per trade in-sample and -14 out-of-sample, and time-series momentum 240 and -13. The historical edge of these rules is largely absent since 2008 under realistic fills and costs, which bounds what any component can add.

#### I. Bias Audit

Table VII lists each threat to validity named in the design, the test applied, and its outcome.

### VII. DISCUSSION

#### A. The Taxonomy Predicts the Variance, Not the Sign

Narrowing components, which can only remove or re-weight decisions the rule already makes, produced the smallest and most consistently positive effects in both studies, and in

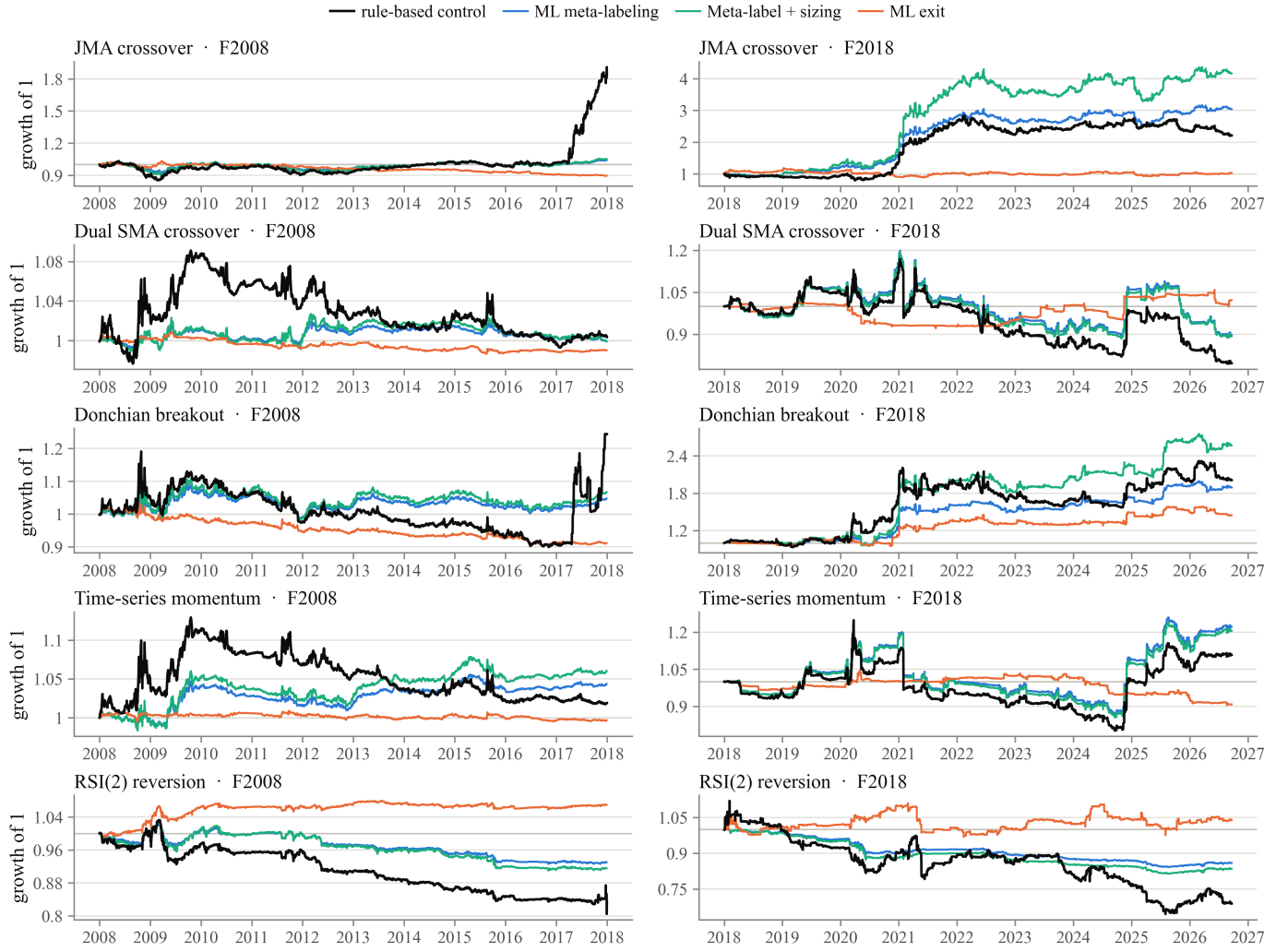


Fig. 11. All-market equity curves (growth of one unit of capital) in each out-of-sample window: the rule-based control against meta-labeling, meta-label + sizing, and ML exit.

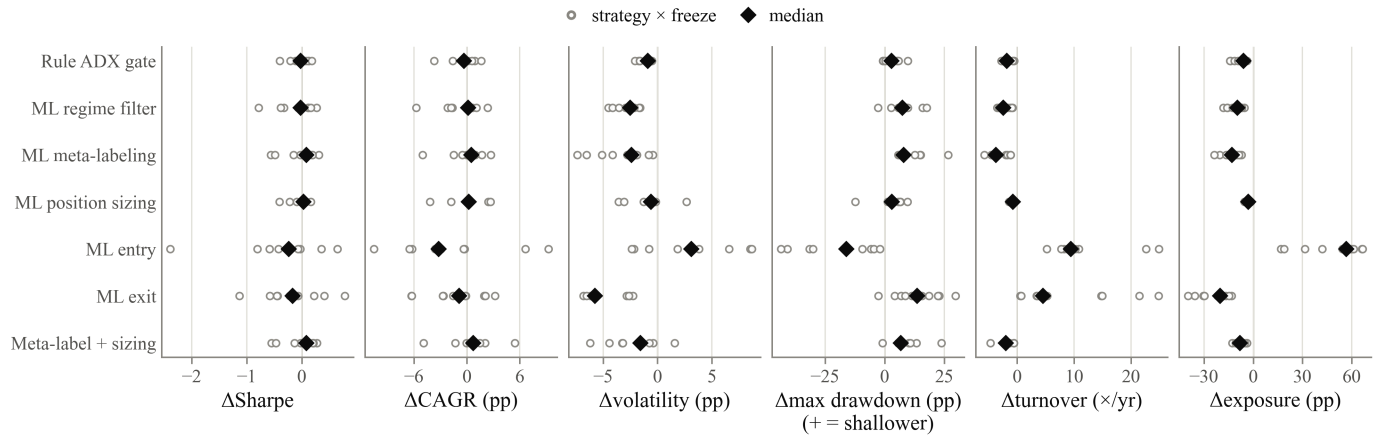


Fig. 12. Change in the all-market risk profile, variant minus control, for each strategy × freeze cell (open circles) and its median (diamond). Drawdown changes track exposure changes: components that trade less have shallower drawdowns.

Study 2 not a single significantly negative one. Replacing components produced the largest effects in both directions. The ex-ante hypothesis of Section III-C is therefore supported.

What the taxonomy does not predict is *which* direction a replacing component takes; that depends on how well the replaced mechanism fits the strategy. A learned continuation

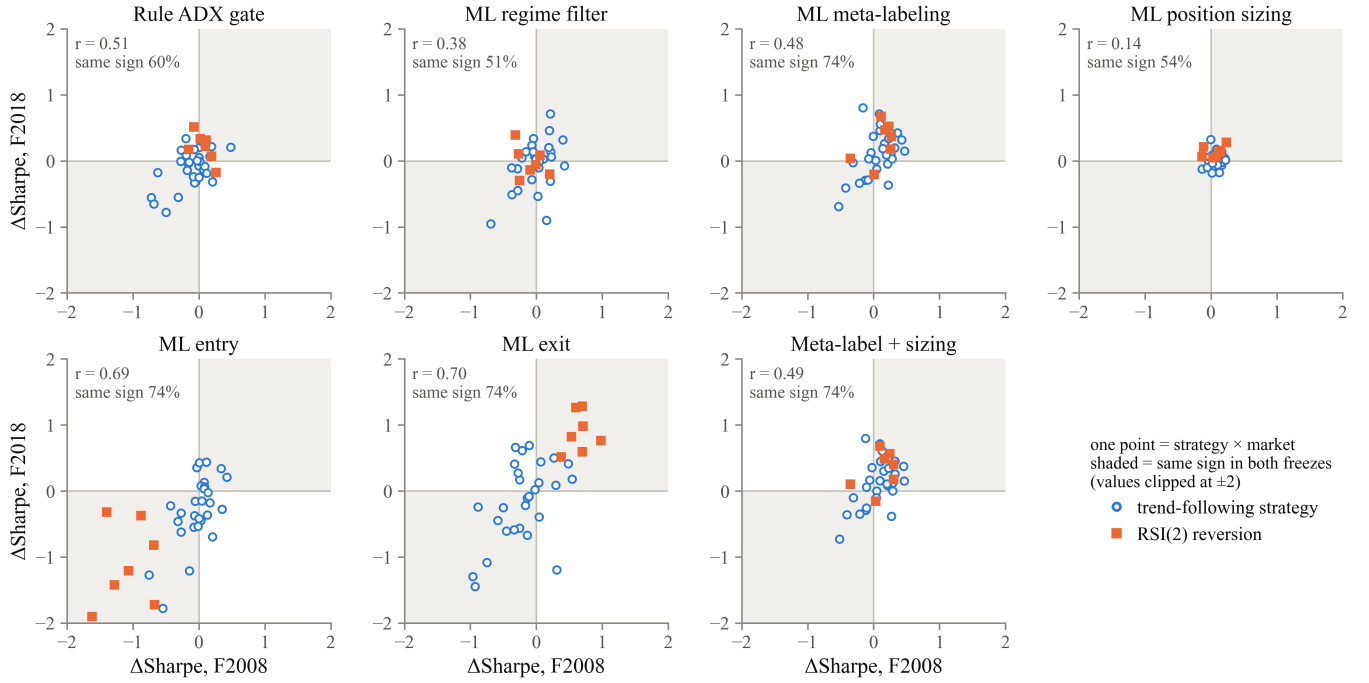


Fig. 13. Replication at a second freeze: market-level  $\Delta$ Sharpe in F2008 against F2018 for each strategy  $\times$  market cell, by component, with the cross-freeze correlation  $r$  and sign-agreement rate.

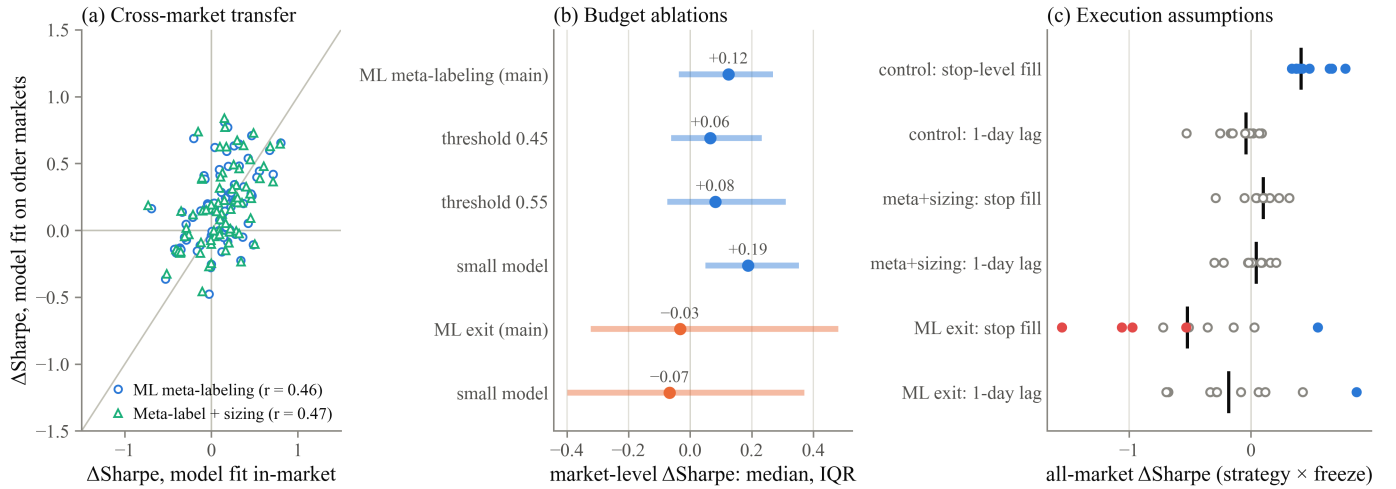


Fig. 14. Robustness. (a) Cross-market transfer:  $\Delta$ Sharpe of models fit on the other markets against models fit in-market. (b) Budget ablations: median and interquartile range of market-level  $\Delta$ Sharpe. (c) Execution: the control under stop-level fills or a one-day lag against the realistic control, and each ML variant against the control under the same execution.

exit is redundant for trend-followers, whose trailing stops already let winners run, but it repairs a genuine weakness of a mean-reversion rule whose fixed profit target truncates them. Symmetrically, a learned entry that finds trends is a poor substitute for a rule designed to fade them. MBCA’s practical output is therefore a question to ask of each insertion point: *does the learned model address a known weakness of the rule it replaces?*

### B. What Study 1 Got Right and Wrong

Study 2 confirms Study 1’s central conclusion—no component reliably and significantly improves a trend-following

control at this budget—and its taxonomy-level reading. It revises three specifics. ML exit is not intrinsically harmful: its Study 1 losses were inflated by optimistic stop fills and are specific to trend-following rules. The meta-label + sizing pairing, pre-registered in Study 2 as an out-of-sample test of a post hoc choice, is positive in most cells but significant in only 1 of 74. And the forex-specific claims of Study 1 do not survive a change of data vendor.

### C. Why the Effects Are Small

The controls themselves have little out-of-sample edge (Fig. 18), and a matched budget of eight generic price features

TABLE VII  
BIAS AUDIT: EACH THREAT TO VALIDITY, THE TEST APPLIED, AND ITS OUTCOME.

| Threat            | Test applied                                                                                                                                   | Outcome                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Look-ahead        | Truncation invariance of every feature and every strategy signal (unit tests); regimes use $t - 1$ data                                        | pass (tests/test_core.py, tests/test_study2.py)                  |
| Data leakage      | All models fit once before the freeze; scrambling all post-freeze data leaves every model bit-identical; labels straddling the freeze excluded | pass (tests)                                                     |
| Survivorship      | Indices/ETFs/futures/FX analysed separately from current-constituent stocks and crypto                                                         | median $\Delta$ SR meta-label+sizing: free +0.11 vs. prone +0.14 |
| Selection         | Universe, strategies, components, windows, statistics pre-registered in git before data existed; every variant reported                        | study.toml first committed in 8d9f50f, before all data/results   |
| Overfitting       | Walk-forward parameter selection; deflated Sharpe; IS vs. OOS trade-return decay; two independent freezes                                      | Fig. 18; freeze agreement in Table V                             |
| Multiple testing  | BH-FDR and Holm within families (528 market-level main tests)                                                                                  | raw $p < 0.05$ : 99; BH: 8 gains / 10 losses; Holm: 16           |
| Execution realism | Close-fill of breached stops (default), 1-day entry lag, cost $\times 0-4$ , per-market costs                                                  | Fig. 14                                                          |
| Data errors       | Two independent sources (FRED) + browser spot-check (MarketWatch)                                                                              | 5-day return corr. 0.82–1.00; spot-check max diff 0.00           |

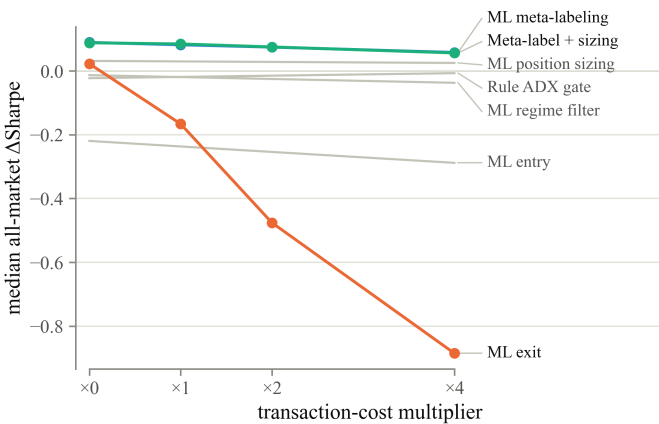


Fig. 15. Median all-market  $\Delta$ Sharpe by component as transaction costs are scaled. The meta-labeling and meta-label + sizing lines nearly coincide.

gives a learned component little information that the rule does not already use. Where the rule has a structural weakness, such as the RSI(2) exit, ML finds it; where it does not, the best a narrowing component can do is trade a little less. This is consistent with the data-snooping literature [4], [12]: once rules are fixed in advance and evaluated under realistic costs, the room for improvement is small.

#### D. Practical Implications

Fig. 19 condenses the evidence into a decision guide for a practitioner with a rule-based system. Prefer narrowing components, and expect them to reduce turnover and exposure more than to raise the Sharpe ratio. Use a replacing component only where the replaced rule has an identifiable weakness, and test it under realistic fills and costs. Evaluate every variant with conservative execution before attributing anything to ML, because execution assumptions move the Sharpe ratio more than any component. Finally, require replication at a second, independent freeze date before acting on a single-window improvement.

### VIII. LIMITATIONS

*One budget.* Every conclusion concerns eight backward-looking price features and one gradient-boosting configuration.

The ablations show that the conclusions are stable to the decision threshold and to a much smaller model, but richer information—fundamentals, order flow, cross-sectional or alternative data—could change them. MBCA is the protocol for testing that fairly, not a claim that the space of learned components is exhausted.

*Daily closes only.* Signals, stops, and fills use daily closes, so intraday stop hits are not modelled. The close-fill convention is conservative relative to stop-level fills, but overnight gaps and liquidity at the open are not simulated.

*Data.* Study 2 relies on one primary vendor, cross-checked against FRED and MarketWatch. Continuous futures are unadjusted front-month rolls; single stocks are current large capitalizations and the crypto series are surviving coins. These survivorship-prone markets are analysed separately and do not drive the conclusions, and one planned symbol was unavailable.

*Portfolio construction.* Equal weighting across instruments with daily mark-to-market accounting is transparent but simple. Volatility targeting or risk parity would change the absolute numbers, though not the paired variant-versus-control design.

*Statistics.* The block bootstrap assumes stationarity within each window, and FDR control is applied within pre-defined families rather than across the whole study. Most market-level effects are fragile to dropping a single instrument; only the significant ones are shown to be robust to it.

*Scope.* The evidence covers five rule families on daily data. MBCA itself is domain agnostic, but we have not applied it outside trading.

### IX. CONCLUSION

Matched-budget component attribution asks where, specifically, a learned component adds value inside a rule-based system, holding everything except the insertion point fixed. Applied to one strategy in three markets (Study 1) and then, pre-registered, to five historically documented strategies on 153 instruments across 8 markets and two independent freezes (Study 2), it gives a consistent answer. At a matched budget of generic price features, ML rarely changes a rule-based strategy significantly (8 significant gains and 10 significant

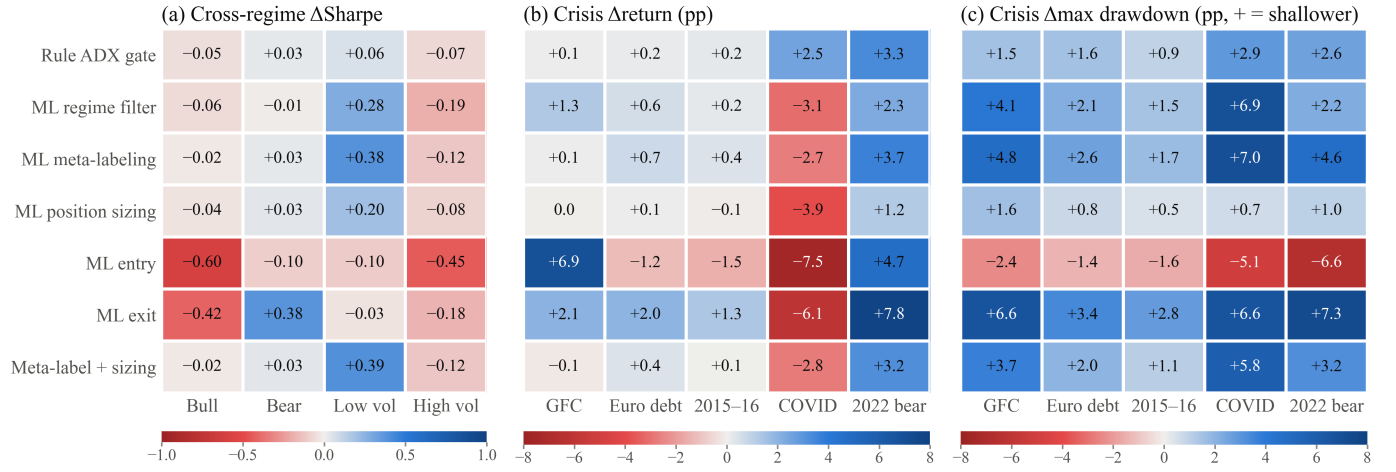


Fig. 16. Regimes and crises. (a) Mean all-market  $\Delta$ Sharpe within bull, bear, low-, and high-volatility regimes defined from information at  $t - 1$ . (b) Change in cumulative return and (c) change in maximum drawdown inside the five pre-registered crisis windows, averaged over strategies (percentage points).

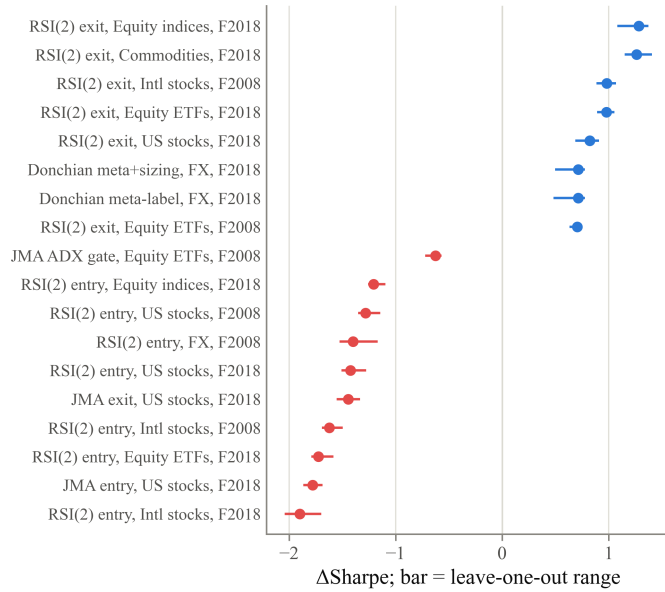


Fig. 17. Leave-one-instrument-out robustness of the 8+10 significant market-level effects: the full-sample  $\Delta$ Sharpe (dot) and its range when any one instrument is dropped (bar). No effect changes sign.

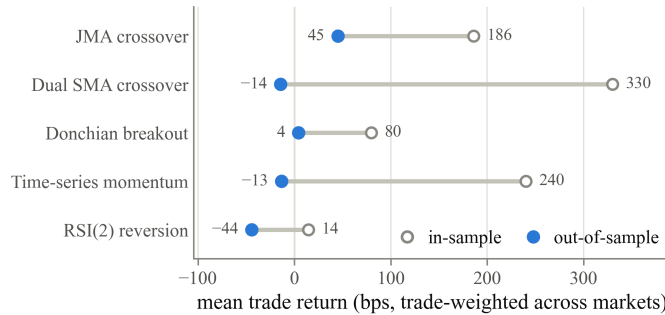


Fig. 18. Mean trade return of the walk-forward-selected controls in-sample and out-of-sample, trade-weighted across markets.

losses in 528 tests). Narrowing components are small, mostly

positive, transferable, and safe. Replacing components are high-variance and help only when the replaced rule has a structural weakness, as ML exit does for RSI(2) mean reversion. And realistic execution matters more than any component. Because every number in this paper is generated from one pre-registered configuration by the released pipeline, each of these claims can be re-derived—or overturned—by re-running it.

#### DATA AND CODE AVAILABILITY

The code, the raw and cleaned data with their provenance manifests, the trades of the reported run, and all analysis outputs are available at [https://github.com/Cannonbold2412/ML\\_Component\\_Attribution](https://github.com/Cannonbold2412/ML_Component_Attribution). The per-configuration results summarized in the figures are provided there as supplementary data: every Study 1 configuration in `results/paper/`, the configuration-by-configuration comparison with the re-implementation in `results/replication/`, and every Study 2 test, portfolio, regime, crisis, and jackknife result in `results/study2/analysis/`.

#### APPENDIX A

##### REPRODUCIBILITY AND PROVENANCE

The chain from raw bytes to every number in this paper has five steps, each a command of the released package:

- 1) `study.toml`: the pre-registered design and the only configuration the code reads.
- 2) `mbca study fetch, clean, and validate`: raw API responses with URL, UTC timestamp, and SHA-256 hash; cleaned series with per-rule row counts and hashes; and cross-source validation reports.
- 3) `mbca study run`: refuses to start with uncommitted code, configuration, or data, and writes every out-of-sample trade together with a run manifest (git commit, configuration and data hashes, package versions, thread settings, and per-task timings).
- 4) `mbca study analyze`: every statistic, written to `results/study2/analysis/`.

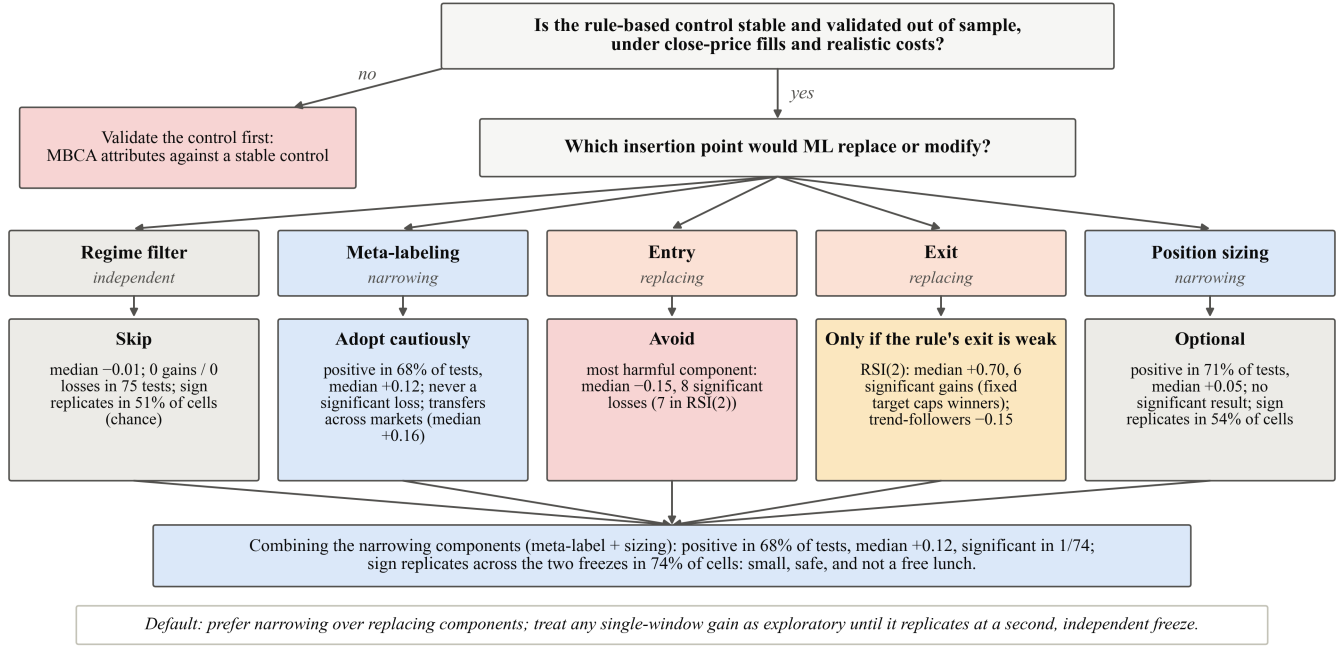


Fig. 19. Decision guide for adding ML to a rule-based system, condensing the evidence of both studies per insertion point. Every number in the boxes is computed from the Study 2 test results by the pipeline.

5) mbca study paper: every table, figure, and in-text number of this manuscript, which contains no hand-typed result.

Human decisions and deviations are recorded in an experiment log, and the version history shows that the configuration file precedes all Study 2 data and results. The test suite enforces every methodological invariant: no look-ahead in features, strategies, or regime labels; freeze integrity (scrambling all post-freeze data leaves every fitted model bit-for-bit identical); one insertion point per component; fill realism; mark-to-market accounting; and the size and power of the bootstrap. The Study 1 results are the original study's output files, shipped verbatim.

## APPENDIX B

### STUDY 1 STATISTICAL CONVENTIONS

Study 1 retains the conventions of the original study so that its results are reported as they were produced. Each variant's trades are converted to a realized-at-exit daily series, and each daily series is resampled independently and i.i.d. (2,000 resamples); the reported CI is that of the difference in *per-day* Sharpe ratios. The CI bounds are therefore  $\sqrt{252} \approx 15.9$  times smaller than the annualized  $\Delta$ Sharpe reported beside them, and the forest plot of Fig. 4 rescales them by  $\sqrt{252}$  so that point and interval share one axis. Because this test ignores both the pairing of the two series and their serial dependence, Study 2 replaces it with the paired circular-block bootstrap of Section V-F and reports mark-to-market rather than realized-at-exit portfolios.

## REFERENCES

- [1] S. Gu, B. Kelly, and D. Xiu, "Empirical asset pricing via machine learning," *Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020.
- [2] M. López de Prado, *Advances in Financial Machine Learning*. Wiley, 2018.
- [3] J. F. Joubert, "Meta-labeling: Theory and framework," *Journal of Financial Data Science*, vol. 4, no. 3, pp. 31–44, 2022.
- [4] R. Sullivan, A. Timmermann, and H. White, "Data-snooping, technical trading rule performance, and the bootstrap," *Journal of Finance*, vol. 54, no. 5, pp. 1647–1691, 1999.
- [5] H. White, "A reality check for data snooping," *Econometrica*, vol. 68, no. 5, pp. 1097–1126, 2000.
- [6] C. R. Harvey, Y. Liu, and H. Zhu, "... and the cross-section of expected returns," *Review of Financial Studies*, vol. 29, no. 1, pp. 5–68, 2016.
- [7] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018.
- [8] V. N. K. Pillai, A. Ajith, and S. K. J., "Generating alpha: A hybrid AI-driven trading system integrating technical analysis, machine learning and financial sentiment for regime-adaptive equity strategies," 2026, arXiv:2601.19504.
- [9] Y. Du, "Deep learning enhanced multi-day turnover quantitative trading algorithm for Chinese A-Share market," 2025, arXiv:2506.06356.
- [10] A. Moreira and T. Muir, "Volatility-managed portfolios," *Journal of Finance*, vol. 72, no. 4, pp. 1611–1644, 2017.
- [11] W. Brock, J. Lakonishok, and B. LeBaron, "Simple technical trading rules and the stochastic properties of stock returns," *Journal of Finance*, vol. 47, no. 5, pp. 1731–1764, 1992.
- [12] C.-H. Park and S. H. Irwin, "What do we know about the profitability of technical analysis?" *Journal of Economic Surveys*, vol. 21, no. 4, pp. 786–826, 2007.
- [13] T. J. Moskowitz, Y. H. Ooi, and L. H. Pedersen, "Time series momentum," *Journal of Financial Economics*, vol. 104, no. 2, pp. 228–250, 2012.
- [14] B. Hurst, Y. H. Ooi, and L. H. Pedersen, "A century of evidence on trend-following investing," *Journal of Portfolio Management*, vol. 44, no. 1, pp. 15–29, 2017.
- [15] P. R. Hansen, "A test for superior predictive ability," *Journal of Business & Economic Statistics*, vol. 23, no. 4, pp. 365–380, 2005.
- [16] A. W. Lo, "The statistics of Sharpe ratios," *Financial Analysts Journal*, vol. 58, no. 4, pp. 36–52, 2002.

- [17] O. Ledoit and M. Wolf, “Robust performance hypothesis testing with the Sharpe ratio,” *Journal of Empirical Finance*, vol. 15, no. 5, pp. 850–859, 2008.
- [18] D. N. Politis and J. P. Romano, “A circular block-resampling procedure for stationary data,” in *Exploring the Limits of Bootstrap*, R. LePage and L. Billard, Eds. Wiley, 1992, pp. 263–270.
- [19] D. H. Bailey and M. López de Prado, “The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality,” *Journal of Portfolio Management*, vol. 40, no. 5, pp. 94–107, 2014.
- [20] D. H. Bailey, J. M. Borwein, M. López de Prado, and Q. J. Zhu, “The probability of backtest overfitting,” *Journal of Computational Finance*, vol. 20, no. 4, pp. 39–69, 2017.
- [21] Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate: A practical and powerful approach to multiple testing,” *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.
- [22] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [23] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems 30*, 2017, pp. 3146–3154.
- [24] C. M. Faith, *Way of the Turtle*. McGraw-Hill, 2007.
- [25] L. A. Connors and C. Alvarez, *Short Term Trading Strategies That Work*. TradingMarkets Publishing Group, 2008.